1.6 Solving Linear Systems Using Technology

Name _____ Period ____ Date _____

Algebra II 1.6 · Student sheet

I can…

- write an augmented matrix for a system and state its dimensions.
- use technology to solve a system of linear equations.
- use technology to solve real-life problems.

Vocabulary · fill in the missing word

Matrix — a _____ arrangement of numbers in rows and columns

Dimensions — m × n: m rows by n _____

Augmented matrix — the _____ and constants of a system, with each equation as a row

Reduced row-echelon form (rref) — the form _____ gives, with 1s on the diagonal and 0s elsewhere; the last column is the solution

Part 1 · Calculator walk-through

Furniture store system: s + l = 1250, s + 2c = 1400, s + l + c = 1550

Augmented matrix (columns s, l, c | constant):

rref result:

Sofa $_____ Love seat $_____ Chair $_____

Part 2 · Notes and examples (fill in each note, then work the example)

Dimensions: Dimensions are _____ × _____.

Augmented matrix: Write each equation in _____ form first, and use _____ for a missing variable.

Example 1

Write an augmented matrix for the system. Then state its dimensions.

$2x - y + 3z = 1$

$x + 4z = -1$

$-3x + 2y - z = -4$

Reading rref: In rref, the solution is in the _____ column.

Example 2

Use technology to solve the system in Example 1.

Example 3

Write an augmented matrix for the system, then solve it with technology.

$3x + y = 2z + 6$

$y - 4 = x$

$z = x$

Example 4

A shop sells T-shirts for $12, hats for $8, and mugs for $5. One day it sold 40 items for $360, with twice as many T-shirts as mugs. How many of each did it sell?

Part 3 · Partner practice

Partner 1

Write an augmented matrix and give its dimensions.

$x + y + z = 4$

$2x - z = 3$

$y + 3z = 8$

Partner 2

Use technology to solve.

$x + 2y - z = 3$

$2x - y + z = 7$

$x + y + z = 6$